

Gravitational Lensing and the Lensing of Gravitational Waves

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Gravitational lensing is most commonly thought about in terms of electromagnetic radiation, a phenomenon that can be described by geometric optics. However, gravitational radiation in the form of gravitational waves can be lensed just the same. This lensing can usually be described by simple geometric optics too, although some cases require wave optics. Lensing of both types can also help with our understanding of dark matter and objects in our galaxy's halo. However, analysis of lensed gravitational waves requires them to be detectable. Although there have been some promising candidates for lensed gravitational waves, the low probabilities for detecting such an event rule it out. With next generation detectors on the horizon, there should soon be a plethora of data on lensed gravitational waves with which we can explore the composition of the universe.

I. INTRODUCTION

Despite the apparent complexity of gravitational lensing of electromagnetic radiation, under reasonable approximations, we find that simple geometric optics is sufficient to accurately describe this phenomenon. Surprisingly, for the lensing of gravitational radiation in the form of gravitational waves, we can often apply the same approximations of simple geometric optics. However, in some cases, the approximations of geometric optics break down and more complex wave optics are required.

With these models, we can find properties like the amplification of a source due to lensing. Furthermore, we can explore the composition of galaxies. Electromagnetic lensing helps us look for dark compact objects in our own galaxy's halo, while the lensing of gravitational waves can uncover aspects about the dark matter substructure of galaxies.

Although lensed gravitational waves have not yet been detected, we can learn about phenomena from their absence, such as limits on the abundance and size of heavy compact dark matter objects. Even so, next generation gravitational wave detectors should provide us with a plenty of data characterizing this phenomenon, opening another window into understanding the structure of the universe around us.

II. GRAVITATIONAL LENSING

Gravitational lensing of electromagnetic waves is surprising simple to describe, following geometric optics to excellent approximation. With knowledge about the deflection of light from a body of mass M , we can easily calculate the deflection angle α in terms of the impact parameter b shown in Figure 1. We can start with our effective potential for photon orbits and the expression for b^2 in terms of V_{eff} and $dr/d\lambda$ where λ is an affine parameter.

$$V_{\text{eff}}(r) = \frac{1}{r^2} \left(1 - \frac{2M}{r} \right) \quad (1)$$

$$b^2 = \frac{1}{\ell} \left(\frac{dr}{d\lambda} \right)^2 + V_{\text{eff}}(r) \quad (2)$$

Here, $b^2 \equiv \ell/e$, e is the total energy per unit rest mass given by $e = \left(1 - \frac{2M}{r} \right) \frac{dt}{d\lambda}$, and ℓ is the angular momentum per unit rest mass given by $\ell = r^2 \frac{d\phi}{d\lambda}$. Equation 2 and the equation for ℓ can be rearranged to solve for the derivatives with respect to λ , giving $\frac{d\phi}{d\lambda} = \ell/r^2$ and $\frac{dr}{d\lambda} = \ell \left(\frac{1}{b^2} - V_{\text{eff}} \right)^{1/2}$. We can divide the two equations, giving an equation for $\frac{d\phi}{dr}$.

$$\frac{d\phi}{dr} = \frac{1}{r^2} \left[\frac{1}{b^2} - V_{\text{eff}}(r) \right]^{-1/2} \quad (3)$$

This can then be turned into the integral expression

$$\Delta\phi = 2 \int_{r_1}^{\infty} \frac{dr}{r^2} \left[\frac{1}{b^2} - \frac{1}{r^2} \left(1 - \frac{2M}{r} \right) \right]^{-1/2} \quad (4)$$

with the factor of 2 stemming from the inward and outward path of the light being symmetric. The turning point of path, r_1 , occurs when $1/b^2 = V_{\text{eff}}(r_1)$, and the bracket in the previous integral expression goes to zero. Then by defining ω by $r = b/\omega$, we can rewrite the integral expression as

$$\Delta\phi = 2 \int_0^{\omega_1} d\omega \left[1 - \omega^2 \left(1 - \frac{2M\omega}{b} \right) \right]^{-1/2} \quad (5)$$

where ω_1 is the value of ω for which the bracket within the integral goes to 0. To evaluate this integral, we can expand it into the form

$$\Delta\phi = 2 \int_0^{\omega_1} d\omega \left(1 - \frac{2M\omega}{b}\right)^{-1/2} \cdot \left[\left(1 - \frac{2M\omega}{b}\right)^{-1} - \omega^2 \right]^{-1/2} \quad (6)$$

which simplifies down to

$$\Delta\phi = 2 \int_0^{\omega_1} d\omega \frac{1 + (M/b)\omega}{[1 + (2M/b)\omega - \omega^2]^{1/2}} \quad (7)$$

Evaluating this integral, we find

$$\Delta\phi \approx \pi + \frac{4M}{b} \quad (8)$$

By looking at Figure 1, we see that the deflection angle α is given by

$$\alpha = \Delta\phi - \pi = \frac{4M}{b} \quad (9)$$

$$= \frac{4GM}{c^2 b} (\text{inserting factors of } G \text{ and } c) = \frac{2R_s}{b} \quad (10)$$

where R_s is the Schwarzschild radius of $2GM/c^2$ [1]. This is valid only for small M/b , meaning the impact parameter must be much larger than the size of the lens. As seen in Figure 1, we consider sharp angles when describing lensing even though the path is actually curved. This is because the distance within which the deflection occurs is much less than the distance over which the light propagates, so the deflection can be approximated as a sharp angles to high accuracy.

Additionally, in reality, the angles within Figure 1 are very small, so the small angle approximation can be used. Using these helpful approximations, we find

$$\theta D_S = \beta D_S + \alpha D_{LS} \quad (11)$$

Then we can substitute equation 10 into equation 11.

$$\theta D_S = \beta D_S + \frac{2R_s}{b} D_{LS} \quad (12)$$

$$\Rightarrow \theta = \beta + \frac{2R_s}{b} \frac{D_{LS}}{D_S} \quad (13)$$

We can then define the *Einstein angle* which determines the angular scale of the lensing.

$$\theta_E \equiv \left[2R_s \left(\frac{D_{LS}}{D_S D_L} \right) \right]^{1/2} \quad (14)$$

Rewriting equation 13,

$$\Rightarrow \theta = \beta + \frac{\theta_E^2 D_L}{b} \quad (15)$$

The sense of scale given by the Einstein angle is important because it shows that many cases of gravitational lensing are at too small to be resolved by telescopes. For the case of intragalactic lensing with a stellar mass, $M \sim M_\odot$, $R_s \sim 1$ km, and $D_L \sim D_S \sim D_{LS} \sim 10^{17}$ km, we find $\theta_E \sim 10^{-3''}$ which is below modern telescope resolution [1].

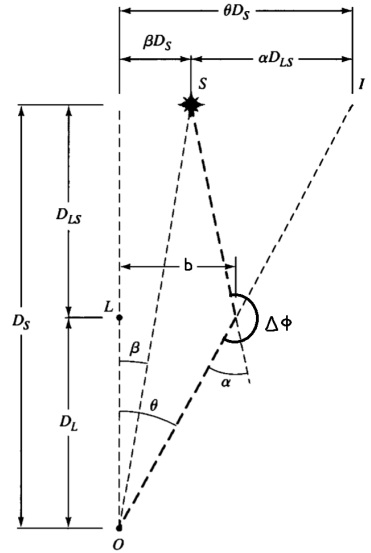


Figure 1. Diagram of gravitational lensing using geometric optics. O is the observer, L is the lens, and S is the source. D_L is the distance from the observer to the lens, D_{LS} is the distance from the lens to the source, and D_S is the distance from the observer to the source. α is the deflection angle, β is the angle between the lens and the source, and θ is the angle between the lens and the image. $\Delta\phi$ is the total angle swept out by the light ray [1].

A. Microlensing

Despite the many cases of lensing being below the resolution of modern telescopes, this does not mean that lensing cannot be used to great affect. Much can be learned through the use of microlensing which causes

fluctuations in the brightness of a source. We can explore this amplification through the use of the solid angle spanned by the image of a source shown in figure 2. Using the fact that a solid angle $\Omega = S/r^2$ where S is the area subtended by the solid angle and r is the radius of the sphere, we find that

$$d\Omega = dS/r^2 \quad (16)$$

From Figure 2 we can see that this solid angle is smaller from the perspective of the observer for a lensed image compared to a non-lensed one. Since the amplification of the intensity of image is directly proportional to the solid angle it subtends, we can find an amplification factor, A , due to gravitational lensing. Using $d\Omega_S$ as the differential solid angle of the unlensed source and $d\Omega_i$ as differential solid angle of the image, we find that

$$A_i = \frac{d\Omega_i}{d\Omega_S} = \left| \frac{\theta_i d\phi d\theta_i}{\beta d\phi d\beta} \right| \quad (17)$$

where A_i is the amplification factor for the i^{th} image, θ_i is the angle between the lens and the i^{th} image, and β and $d\phi$ are defined in Figure 2.

In the case of Figure 2, there are two images so the amplification must take both images into account. First, we can find that

$$\theta_{1,2} = \frac{1}{2} \left[\beta \pm \sqrt{\left(\frac{2R_E}{D_L} \right)^2 + \beta^2} \right] \quad (18)$$

where $R_E \equiv 2\sqrt{\frac{GmD_L D_{LS}}{c^2 D_S}}$. We can also define

$$\Delta\theta \equiv \theta_1 - \theta_2 = \sqrt{\left(\frac{2R_E}{D_L} \right)^2 + \beta^2} \quad (19)$$

Then, from equation 18, we find that $d\theta/d\beta = \frac{1}{2}(1 \pm \beta/\Delta\theta)$. Substituting back into equation 17 and taking both images from Figure 2 into account, we find

$$A_{tot} \equiv A_1 + A_2 = \frac{1}{2} \left[\frac{\beta}{\Delta\theta} + \frac{\Delta\theta}{\beta} \right] \quad (20)$$

Using $u = \frac{\beta D_L}{R_E}$, we find that equation 20 can be rewritten as

$$A = \frac{u^2 + 2}{u\sqrt{u^2 + 4}} \quad (21)$$

with the derivation found in the appendix, section VII. With this expression for the amplification, we can explore u , expressing it as a function of time. If we consider a lens with velocity v moving between the source and the observer, we find that

$$u^2(t) = u_{min}^2 + \left[\frac{t - t_0}{T} \right]^2 \quad (22)$$

where $T = R_E/v$, $u_{min} = d/R_E$, and d is the minimum distance between the lens and the source. We can plug this time-dependent expression for u into equation 21. We find a function that look like those in Figure 3, with a maximum around t_0 and larger maxima at smaller u_{min} . Thus for smaller u_{min} , the amplification is much larger [2].

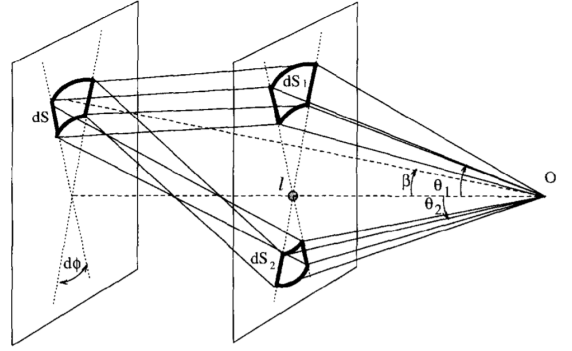


Figure 2. Diagram of the two images resulting from lensing. O is the observer and l is the lens, with β as the angle between the lens and the source. θ_1 and θ_2 are the angles between the lens and their respective images [2].

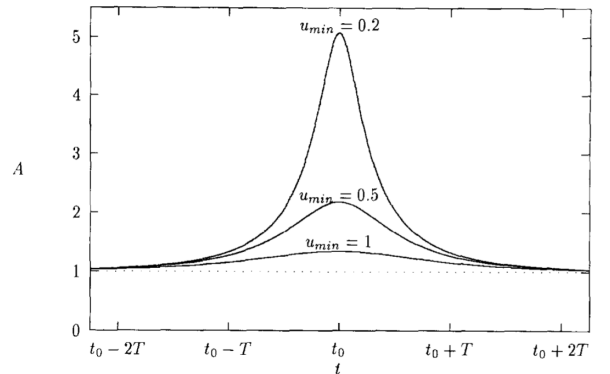


Figure 3. Graph of the time evolution of the amplification of the source for different values of u_{min} [2].

III. GRAVITATIONAL WAVE LENSING

A. Approximations

Although the lensing of gravitational waves may seem like a different phenomenon, it is still the lensing of radiation. Thus the gravitational lensing of gravitational waves can be described by the geometric optics described above in many cases. However, this is not true for all situations. As shown in Figure 2, there can be multiple images for lensed radiation. There is a time delay between the arrival of the images since they travel along different paths. For the lensing of gravitational waves, if the time delay of a signal is less than or on the order of the signal's period, diffraction starts to have significant effects. For a lens of mass M_L , the above condition is met roughly for $M_L \lesssim 10^3 M_\odot$. In terms of LIGO which looks at gravitational wave in the $10 - 10^4$ Hz range, lenses of this mass correspond microlensing [3].

This is not the only approximation in geometric optics that often breaks down for gravitational wave lensing. In electromagnetic lensing, the wavelength of the electromagnetic radiation is much less than the size of the lens. However, for low frequency gravitational waves, the wavelength can be upwards of 10^{10} km [4]. In this case, geometric optics cannot be used, as the useful approximations that geometric optics allows us to make are no longer reasonable. Roughly, the geometric approximation breaks down for a gravitational wave of wavelength $\lambda \gtrsim GM/c^2$ [5]. When the geometric approximations are broken, we must begin to consider wave optics rather than geometric optics.

B. Wave Optics

To illustrate the use of wave optics, we can take the case of waves from a point source. The spacetime metric is taken to be

$$ds^2 = -(1 + 2U)dt^2 + (1 - 2U)d\vec{r}^2 \quad (23)$$

where we have the Minkowski metric with a small perturbation from the gravitational potential of the lens. The propagation equation of a wave $\phi(\vec{r}, t) = \tilde{\phi}(\vec{r})e^{-i\omega t}$ is the scalar wave equation

$$(\nabla^2 + \omega^2)\tilde{\phi} = 4\omega^2 U \tilde{\phi} \quad (24)$$

Using spherical coordinate (r, θ, φ) with the origin at the source and the polar axis pointing at the lens, we can set the observer to be at $\vec{r}_0 = (r_0, \theta_0, \varphi_0)$. Then the amplification is $A(\vec{r}) = \tilde{\phi}(\vec{r})/\tilde{\phi}_0(r)$. Equation 24 can then be rewritten as

$$\frac{\partial^2 A}{\partial r^2} + 2i\omega \frac{\partial A}{\partial r} + \frac{1}{r^2} \nabla_\theta^2 A = 4\omega^2 U A \quad (25)$$

where $\nabla_\theta^2 = \partial^2/\partial\theta^2 + \theta^{-1}\partial/\partial\theta + \theta^{-2}\partial^2/\partial\varphi^2$. If we assume that the scale on which A varies is much greater than the wavelength, we have $\omega/|\partial \ln A/\partial r| \gg 1$ and the first term of equation 25 can be neglected. Then we are left with a Schrödinger-like equation. We can then find the corresponding Lagrangian.

$$L(r, \theta, \dot{\theta}) = \omega \left[\frac{1}{2} r^2 |\dot{\theta}|^2 - 2U(r, \theta) \right] \quad (26)$$

with $\dot{\theta} = d\theta/dr$. Using the path integration formulation of quantum mechanics, we find

$$A(\vec{r}_0) = \int \mathcal{D}\theta(r) \exp \left\{ i \int_0^{r_0} dr L[r, \theta(r), \dot{\theta}(r)] \right\} \quad (27)$$

We can in then interpret this expression in three steps. First, choose $\theta(r)$ as the path from the source at the origin to an observer at $\vec{r}_0 = (r_0, \theta_0)$. Second, by evaluating the integral within the curly brackets along $\theta(r)$, we find something phase-like as a function of $\theta(r)$. Third, the "phases" are summed for all possible $\theta(r)$. If r_0 is broken up into N steps of size ε such that $\varepsilon = r_0/N$ as shown in Figure 4, we can then specify $\theta(r)$ by $\theta_j = \theta(r_j)$, $j = 1, \dots, N$, on the sphere of radius $r_j = j\varepsilon$. Now to simplify, we can use the thin lens approximation, $\frac{1}{2}\delta(r - r_l)\hat{\psi}(\theta)$ with

$$\hat{\psi}(\theta) = 2 \int_0^{r_0} dr U(r, \theta) \quad (28)$$

to substitute for $U(r, \theta)$ in equations 26 and 27. This becomes

$$A(\vec{r}_0) = \left[\prod_{j=1}^{N-1} \int \frac{d^2\theta_j}{B_j} \right] \cdot \exp \left\{ i\omega \left[\varepsilon \sum_{j=1}^{N-1} \frac{r_j r_{j+1}}{2} \left| \frac{\theta_j - \theta_{j+1}}{\varepsilon} \right|^2 - \hat{\psi}(\theta_l) \right] \right\} \quad (29)$$

where $B_j = 2\pi i\varepsilon/(\omega r_j r_{j+1})$ normalizes A such that $A = 0$ if $\hat{\psi} = 0$. Through some manipulation worked out in the appendix, section VII, this becomes

$$A(\vec{r}_0) = \frac{\omega r_l r_0}{2\pi i r_{l0}} \int d^2\theta_l \cdot \exp \left\{ i\omega \left[\frac{r_l r_0}{2r_{l0}} |\theta_l - \theta_0|^2 - \hat{\psi}(\theta_l) \right] \right\} \quad (30)$$

where r_{l0} is the distance between the lens and the observer [6]. Thus we have found the amplification of the gravitational wave when our geometric optics break down using wave optics.

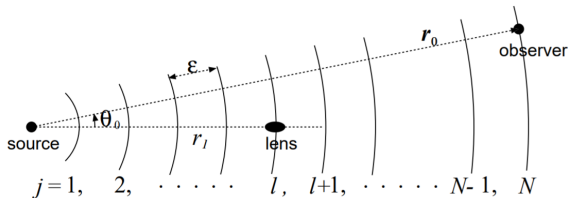


Figure 4. Diagram of the source, observer, and lens, with the distance broken up into N steps of ε . The lens is located at $r_l = l\varepsilon$ [6].

IV. APPLICATIONS TO GRAVITATIONAL LENSING

A. Electromagnetic Lensing and MACHOs

As we saw with microlensing, the brightness of a lensed object changes with time as a lens moves between an observer and a source. This is highly useful in the detection of compact dark objects, as we cannot rely on electromagnetic interactions. A dark object can be used as a lens, allowing us to learn about the abundance of such objects within the outer halo of our own galaxy. Such compact dark objects are called massive compact halo objects, known as MACHOs. If a MACHO within the halo passes between us and a source star in the Large Magellanic Cloud (LMC) outside our galaxy's halo, we will see the brightness of the source change as a function of time, just like in Figure 3. Namely, the brightness will increase and then decrease as the MACHO moves by. The time scale for this fluctuation in brightness, t_{var} , can be estimated as time it takes for the MACHO of velocity v to cross the distance subtended by the Einstein angle, θ_E , defined in equation 14. This gives

$$t_{var} = \frac{\theta_E D_L}{v} \quad (31)$$

where D_L is defined in Figure 1. For a MACHO, we can estimate this as

$$t_{var} = \frac{(10^{-3} \text{ "})(10 \text{ kpc})}{200 \text{ km/s}} \sim 0.2 \text{ yr} \quad (32)$$

a time period that is very measurable [1]. Conversely, and perhaps more practically, we can find the mass of

a lens through this equation. To find the mass of the lens, we can first take

$$\dot{\beta} = \frac{v}{D_L} \approx \frac{\theta_E}{t_{var}} \quad (33)$$

where β is the angle between the lens and the source. Then using equation 14 and the definition for R_S , we find that

$$\begin{aligned} t_{var} &\sim \frac{D_L}{v} \theta_E = \frac{D_L}{v} \left[2R_S \left(\frac{D_{LS}}{D_S D_L} \right) \right]^{1/2} \\ &= \frac{D_L}{v} \left[4M \left(\frac{D_{LS}}{D_S D_L} \right) \right]^{1/2} \end{aligned} \quad (34)$$

Solving for M , we find

$$M = (v \cdot t_{var})^2 \frac{D_S}{4D_{LS}D_L} \quad (35)$$

Thus with measurements for the distances from the observer to the lens and from the lens to the source, we can find the mass of the lens. Then if there is a MACHO that lenses a source in the LMC, we can find its mass and learn more about the galactic structure and its dark constituents.

B. Dark Matter Analysis From Lensed Gravitational Waves

MACHOs are not the only way lensing can help explore matter that does not interact electromagnetically. Gravitational lensing can be used to study dark matter as well. The lensing of electromagnetic signals has been used to help understand structure of dark matter within galaxies since unlike with electromagnetic effects, we can observe dark matter's gravitational effects. However, there is a method for examining dark matter with lensed radiation that has yet to be used. By utilizing quasars as a source, we can observe the lensing of a regular and reliable signal. Since these signals have multiple images separated by a time delay, small perturbations of this time delay between signals can tell us about the dark matter substructure of the galaxy lens. Unfortunately, the current accuracy for measuring these perturbations in the time delay in electromagnetic lensing is inadequate. However, with gravitational waves, events are short, and the time delays between images could be determined within $\sim .1$ s. When considering the perturbation in the time delays, there are five different uncertainties that contribute to σ_{tot} . First is σ_{obs} , consisting of pixel intensities, central velocity dispersion, and point image positions. σ_{LOS} consists of

the fluctuations in the mass density along the line of sight. σ_{cosm} takes into account uncertainty from the cosmological model used to calculate distances. σ_{dm} is perturbations from dark matter in the lens galaxy's subhalo. Then, σ_{arc} encompasses perturbations in the images caused by dark matter halos. Thus we have

$$\sigma_{tot}^2 = \sigma_{obs}^2 + \sigma_{LOS}^2 + \sigma_{cosm}^2 + \sigma_{arc}^2 + \sigma_{dm}^2 \quad (36)$$

When considering the dark matter substructure, the last two uncertainties, σ_{arc} and σ_{dm} , are the uncertainties from dark matter within the lensing galaxy, causing the apparent time delay anomalies [7]. Thus by examining these uncertainties of the time delay, we can learn about the structure of dark matter in far away galaxies.

Though there have not been any detected lensed gravitational waves, we can still learn about the dark matter from their absence. With a lack of lensed gravitational waves, we can put limitations on the abundance of large dark matter objects. In LIGO-Virgo's O1, O2, and O3 observation runs, 50 compact object coalescence events were observed. If all dark matter was composed of compact heavy objects, we would have expected LIGO to have seen 8 lensed events from the gravitational waves. From this, it was found that no more than 40% of dark matter can be composed of compact objects of $M \gtrsim 400M_\odot$ and all dark matter cannot be composed of compact objects of $M > 200M_\odot$. Additionally, with next generation gravitational wave detectors like the Einstein Telescope, we will soon be able to place much stricter bounds on the type of dark matter our universe is composed of. For example, if even 10% of dark matter is composed of compact objects, then almost all gravitational wave detections from the Einstein telescope will be lensed [8].

V. DETECTION OF LENSED GRAVITATIONAL WAVES

Currently, there have been no detections of lensed gravitational waves. There are a handful of events that are consistent with the lensing hypothesis. However, when taking into account selection effects and the low probability of detecting a lensed event, there is not sufficient evidence for gravitational wave lensing at the current time. This is unsurprising, since the ratio of lensed to unlensed events ranges from $\sim 10^{-3}$ to $\sim 10^{-4}$ and LIGO has detected ~ 90 gravitational wave events [9].

To be more specific, we can look at the lensing of binary neutron star (BNS) events and the resulting amplification A from the lensing. It has been found that for BNS events that are gravitationally lensed with $A > 1.02$, the probability of a detection by LIGO is less than 7×10^{-8} with the sensitivity of LIGO's O1 and

O2 runs [10]. Thus it is unsurprising that we have not detected lensed BNS events, and these lensed events are likely to remain undetected in the near future.

Despite the low chance of lensed BNS event detection, the same is not true for detecting lensed stellar binary black hole coalescences. Analysis suggests that we should expect Advanced LIGO to detect $\sim 1\text{yr}^{-1}$ when it is at design sensitivity. This analysis needs to take into account the time delay between multiple images of the lensed events. After all, for an observing run of finite length and a time delay between the arrival of images of the lensed event, there can be cases where one image arrives during the observing run and others do not. It has been found that for typical double lensed events, 90% of systems have time delays less than ~ 1 month. Thus for future gravitational wave detectors, this should not be an issue with observing runs lasting much longer than the lensing time delay. Furthermore, for typical quadruple lensed events with four images, 90% of systems have time delays less than $\sim .4$ months, so time delays for quadruple lensed events are even less of an issue. However, with differing time delays for double lensed and quadruple lensed events, there is room for quadruple lensed events to be over-represented in an observing run as they have a shorter time delay [11]. Nonetheless, we should be able to detect a lensed gravitational wave event in the near future.

VI. CONCLUSION

Gravitational lensing provides us with an application of geometric optics on the largest of scales. Despite the phenomenon seeming complex, under reasonable approximations such as the wavelength of electromagnetic radiation being much smaller than the lens, it can be simplified into a simple diagram of distances and angles. With this wonderful simplification, we can describe numerous parts of gravitational lensing of electromagnetic radiation such as the location of images and the amplification of a signal. Additionally, with techniques like microlensing, we can find the amplification of a signal as a function of time as a lens passes between us and a source. With this technique, we are able to solve for things like the mass of the lens, allowing us to learn about different galactic objects like MACHOs.

These same geometric approximations can often describe the lensing of gravitational radiation as well, given that the limits of our approximations still hold. When this is not the case, we must turn to wave optics, as diffraction of the lensed waves becomes important within our model. Although this is more complicated, we can still solve for the amplification factor of the lensed waves by calling upon Lagrangian mechanics and integration techniques from quantum mechanics. These lensed gravitational waves have applications to explor-

ing dark matter as well. By looking at perturbations in time delays between the images of lensed gravitational waves (something that is much harder with electromagnetic radiation lensing), we can learn about the dark matter substructure within galaxies. Even though no lensed gravitational waves have been detected, we can learn for their absence, placing limits on the size and abundance on compact dark matter objects.

Lastly, we can analyze the detectability of lensed

gravitational waves. It has been found that LIGO has not detected any lensed gravitational wave events, despite some events having many of the required characteristics. With low probabilities of LIGO detecting a lensed event, like the probability of detecting a BNS event in O1 and O2 less than 7×10^{-8} , it was concluded that any promising candidates were not lensed events. However, with next generation detector like the Einstein Telescope and upgrades to LIGO, we are sure to see one of these events in the coming years.

VII. APPENDIX

A. Derivation of equation 21

$$A = \frac{1}{2} \left[\frac{\beta}{\Delta\theta} + \frac{\Delta\theta}{\beta} \right]$$

$$= \frac{1}{2} \left[\frac{\beta}{\sqrt{\left(\frac{2R_E}{D_L}\right)^2 + \beta^2}} + \frac{\sqrt{\left(\frac{2R_E}{D_L}\right)^2 + \beta^2}}{\beta} \right]$$

Working with the right term:

$$\begin{aligned} \frac{\sqrt{\left(\frac{2R_E}{D_L}\right)^2 + \beta^2}}{\beta} &= \frac{R_E}{\beta D_L} \sqrt{\frac{\beta^2 D_L^2}{R_E^2} + 4} \\ &= \frac{\frac{\beta^2 D_L^2}{R_E^2} + 4}{\frac{\beta D_L}{R_E} \sqrt{\frac{\beta^2 D_L^2}{R_E^2} + 4}} = \frac{\beta}{\frac{R_E}{D_L} \sqrt{\frac{\beta^2 D_L^2}{R_E^2} + 4}} \\ &\quad + \frac{4}{\frac{\beta D_L}{R_E} \sqrt{\frac{\beta^2 D_L^2}{R_E^2} + 4}} \end{aligned}$$

If we rewrite the first term of the second term expansion as

$$\frac{\beta}{\sqrt{\beta^2 + \frac{4R_E^2}{D_L^2}}}$$

we can now combine it with the first term:

$$\frac{1}{2} \left[\frac{2\beta}{\sqrt{\beta^2 + \frac{4R_E^2}{D_L^2}}} + \frac{4}{\frac{\beta D_L}{R_E} \sqrt{\frac{\beta^2 D_L^2}{R_E^2} + 4}} \right]$$

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$$\begin{aligned}
&= \frac{1}{2} \left[\frac{\frac{2\beta D_L}{R_E}}{\sqrt{\frac{\beta^2 D_L^2}{R_E^2} + 4}} + \frac{4}{\frac{\beta D_L}{R_E} \sqrt{\frac{\beta^2 D_L^2}{R_E^2} + 4}} \right] \\
&= \frac{1}{2} \left[\frac{\frac{2\beta^2 D_L^2}{R_E^2}}{\frac{\beta D_L}{R_E} \sqrt{\frac{\beta^2 D_L^2}{R_E^2} + 4}} + \frac{4}{\frac{\beta D_L}{R_E} \sqrt{\frac{\beta^2 D_L^2}{R_E^2} + 4}} \right] \\
&= \frac{\frac{\beta^2 D_L^2}{R_E^2} + 2}{\frac{\beta D_L}{R_E} \sqrt{\frac{\beta^2 D_L^2}{R_E^2} + 4}} = \frac{u^2}{u\sqrt{u^2 + 4}}
\end{aligned}$$

B. Derivation of equation 30

All the integrals in equation 29 are Gaussian and result in unity for $j = 1$ to $j = l - 1$. Thus the lower bound on Π and Σ is $j = l$. Then using the identity

$$\begin{aligned}
\sum_{j=l}^{N-1} r_j r_{j+1} |\theta_j - \theta_{j+1}|^2 &= \varepsilon \frac{r_l r_0}{r_{l0}} |\theta_l - \theta_0|^2 \\
&+ \sum_{j=l+1}^{N-1} r_j^2 \frac{r_{l,j+1}}{r_{lj}} |\theta_j - u_{lj}|^2
\end{aligned}$$

where $r_0 = r_N$, $r_{lj} = r_j - r_l$, and $u_{lj} = [r_l \theta_l + (j - l)r_{j+1} \theta_{j+1}] / (jr_{l,j+1})$. Equation 29 then expands to

$$\begin{aligned}
A(\vec{r}_0) &= \int \frac{d^2 \theta_l}{B_l} \exp \left\{ i\omega \left[\frac{r_l r_0}{2r_{l0}} |\theta_l - \theta_0|^2 - \hat{\psi}(\theta_l) \right] \right\} \\
&\cdot \left[\prod_{j=l+1}^{N-1} \int \frac{d^2 \theta_j}{B_j} \right] \exp \left[\frac{i\omega}{2\varepsilon} \sum_{j=l+1}^{N-1} r_j^2 \frac{r_{l,j+1}}{r_{lj}} |\theta_j - u_{lj}|^2 \right]
\end{aligned}$$

The Gaussian integrals can then be evaluated, and the second line of the previous equation gives

$$\prod_{j=l+1}^{N-1} \frac{1}{B_j} \frac{2\pi i \varepsilon r_{lj}}{\omega r_j^2 r_{l,j+1}} = \prod_{j=l+1}^{N-1} \frac{r_{j+1} r_{lj}}{f_j r_{l,j+1}} = \frac{r_0 \varepsilon}{r_{l+1} r_{l0}}$$

Substituting this back in, we get equation 30.

$$\begin{aligned}
A(\vec{r}_0) &= \frac{\omega r_l r_0}{2\pi i r_{l0}} \int d^2 \theta_l \\
&\cdot \exp \left\{ i\omega \left[\frac{r_l r_0}{2r_{l0}} |\theta_l - \theta_0|^2 - \hat{\psi}(\theta_l) \right] \right\}
\end{aligned}$$